
Artificial Intelligence

Assignment 3

Total Points: 5

Knowledge Representation & Reasoning

Due: 17:00, 24 April, 2026

Submission PDF to: <https://f.wps.cn/g/whW70KW9/>

1. Logical Representation (1.5 pts)

1. Translate the following English sentences into FOL. Consider predicates

$$\text{Person}(x), \text{Likes}(x, y), \text{Older}(x, y), \text{Happy}(x), \text{Friend}(x, y), \text{Teacher}(x, y)$$

- (a) Everyone likes John.
 - (b) No one is older than themselves.
 - (c) All friends of John are happy.
 - (d) If someone is a teacher of Tim, then that teacher is older than Tim.
 - (e) There is a person who likes everyone who likes them.
2. Translate each FOL sentence into an English sentence.
- (a) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Friend}(x, y) \wedge \text{Happy}(y)))$
 - (b) $\exists x (\text{Person}(x) \wedge \forall y (\text{Person}(y) \rightarrow \text{Older}(x, y)))$
 - (c) $\forall x (\text{Person}(x) \rightarrow (\text{Friend}(\text{John}, x) \leftrightarrow \text{Likes}(x, \text{John})))$
 - (d) $\exists x (\text{Person}(x) \wedge \text{Likes}(x, \text{Tim}) \wedge \forall y ((\text{Person}(y) \wedge \text{Likes}(y, x)) \rightarrow \text{Happy}(y)))$
 - (e) $\forall x (\text{Person}(x) \rightarrow (\exists y (\text{Teacher}(y, x) \wedge \text{Older}(y, x)) \rightarrow \text{Happy}(x)))$

2. Logical Inference (1.5 pts)

1. Given the following definite clause knowledge base KB, apply SLD resolution to prove that

$$\text{KB} \models \text{can_rent}(\text{Company2}, \text{John}, \text{Car1})$$

Provide the substitutions resulting from unification at each step. Draw the search tree of the SLD resolution process.

Clause	
1	has_license(John)
2	get_insurance(John, Insurance, Company1)
3	above21(John)
4	available(Company2, Car1)
5	company(Company1)
6	company(Company2)
7	$\forall p, x : \text{eligible}(p) \leftarrow \text{has_license}(p) \wedge \text{get_insurance}(p, \text{Insurance}, x) \wedge \text{above21}(p) \wedge \text{company}(x)$
8	$\forall x, p, c : \text{can_rent}(x, p, c) \leftarrow \text{eligible}(p) \wedge \text{available}(x, c) \wedge \text{company}(x)$

3. Bayesian Networks (2 pts)

Consider a Bayesian network containing the following binary variables (Y or N) and CPT:

- G : Game genre is Action
- E : Player is experienced
- D : Game is difficult
- T : Playtime > 10 hours/week
- F : High enjoyment

CPTs:

$$P(G = Y) = 0.6, \quad P(E = Y) = 0.4$$

G	E	Y	$P(D \mid G, E)$			
N	N	N	0.8			
N	N	Y	0.2	D	T	$P(T \mid D)$
N	Y	N	0.5	N	N	0.7
N	Y	Y	0.5	N	Y	0.3
Y	N	N	0.3	Y	N	0.2
Y	N	Y	0.7	Y	Y	0.8
Y	Y	N	0.1			
Y	Y	Y	0.9			

T	E	F	$P(F \mid T, E)$
N	N	N	0.9
N	N	Y	0.1
N	Y	N	0.6
N	Y	Y	0.4
Y	N	N	0.4
Y	N	Y	0.6
Y	Y	N	0.1
Y	Y	Y	0.9

Derive the following probabilities:

- Compute $P(E = Y \mid F = N)$.
- Compute $P(D = Y \mid T = Y, E = Y)$.